Fermions as torsitons: the charged-lepton hierarchy from a single lattice-measurable scale

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In Einstein–Cartan gravity the elementary fermions can be realized as *torsitons*— chiral solitons of the nonlinear Einstein–Cartan–Dirac equation. Within this picture the entire fermion mass hierarchy reduces to one configurational quantity: the sharpness s_T of the dynamically generated core to which the soliton's radial levels couple. The mass of the n-th generation is a wavefunction overlap, exponentially sensitive to s_T ($\mathrm{d}\ln(\mathrm{span})/\mathrm{d}\ln s_T \approx 15$), so a single sharp core spreads the consecutive levels across the observed $\times 3477$ charged-lepton span; the productive window is $s_T \in [0.43, 0.70]$ in units of the confinement scale. We measure s_T non-perturbatively on a dynamical $N_f = 2$ SU(3) lattice. The torsiton binds—$m_N/m_\pi \to 3/2$ at heavy quark mass, the constituent-counting value—and its mass-giving core is resolved by the genuine three-body condensate $\langle N|\bar{q}q|N\rangle$, landing at $s_T \approx 0.45$–0.56, inside the productive window. Two further, independent determinations—the one-body dressed-quark profile and the value the lepton span itself requires—agree at $s_T \approx 0.43$. The mechanism by which a single geometric scale sets the hierarchy is thus confirmed where the Standard Model inserts nine masses by hand; the chiral-limit value and the absolute scale follow from a dedicated campaign.

I. INTRODUCTION

The masses of the elementary fermions are the largest block of unexplained parameters in the Standard Model. Nine charged-fermion masses, spanning a factor of $\sim$3477 from the electron to the tau and five orders of magnitude from the electron to the top, are inserted as Yukawa couplings to the Higgs field, with no dynamical origin and no relation among them; the four mixing angles and the neutrino sector add more. That this list might instead be a *spectrum*—computable, like the lines of hydrogen, from one equation and a few constants—is among the oldest aspirations of unified theory, articulated already in Heisenberg's nonlinear-spinor programme [1].

A concrete route to such a spectrum is offered by gravity itself. In the Einstein–Cartan extension of general relativity [2], the antisymmetric part of the affine connection—the torsion—is sourced algebraically by the Dirac spin current. Because the torsion does not propagate, it can be eliminated, leaving the minimally coupled Dirac Lagrangian augmented by a quartic spin–spin contact term: the nonlinear Einstein–Cartan–Dirac, or Hehl–Datta, equation [3]. This self-interaction is attractive in the axial channel and admits self-bound, localized solutions—solitons. The same torsion-mediated repulsion, at the opposite extreme of density, halts gravitational collapse and replaces the cosmological singularity with a bounce [4, 5]; the torsiton is its particle-scale counterpart, a standing knot of the one gravity–torsion field.

So posed, the torsiton joins a long line of soliton descriptions of matter. Skyrme's topological soliton of the pion field [6], quantized into the nucleon with its static properties computed to tens of percent [7, 8], established that baryons can be solitons of a bosonic order parameter. The chiral-quark-soliton and Nambu–Jona-Lasinio realizations [9–11] sharpened the picture: the fermion's mass is not a bare input but *configurational*—the field energy bound in a position-dependent constituent mass $M(r)$ that the soliton itself digs from the chiral condensate. The torsiton supplies the gravitational origin of that same self-bound chiral core. The wider ambition—masses, and the regularities among them [12], from structure rather than insertion—also underlies the composite-electroweak programmes, from technicolor [13, 14] to its near-conformal "walking" refinements [15, 16]; we acknowledge these as kin, while differing in that here a single torsion field, not a new gauge sector, carries the dynamics.

The contribution of this paper is to show that, in the torsiton picture, the entire charged-lepton hierarchy collapses to *one* quantity with a clean non-perturbative definition—the sharpness of the mass-giving core, measured in units of the confinement scale—and to determine that quantity on a dynamical lattice. We set the scale by the static potential [17, 18] and track the approach to the chiral limit through the Gell-Mann–Oakes–Renner relation [19]. The paper is organized as follows. Section II derives the configurational mass and the exponential lever that turns one geometric scale into the observed span. Section III measures that scale on the dynamical ensemble, by three independent routes. Section IV discusses what is thereby established and what remains, and connects the result to the broader framework. Section V concludes.

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II. THEORY

A. The torsiton

In Einstein–Cartan gravity the connection carries torsion $S^{\lambda}{}_{\mu\nu}$, whose field equation is algebraic and ties it to the Dirac spin tensor. Eliminating the torsion returns the Dirac field to a Riemannian background at the cost of a quartic self-interaction [2, 3],

$$\mathcal{L} \;=\; \tfrac{i}{2}\,\bar\psi\,\gamma^{\mu}\overleftrightarrow{\partial}_{\mu}\psi - m\,\bar\psi\psi - \frac{3}{16}\,\kappa\,\big(\bar\psi\,\gamma_5\gamma^{a}\psi\big)\big(\bar\psi\,\gamma_5\gamma_a\psi\big),\qquad \kappa = 8\pi G, \tag{1}$$

the axial-current–axial-current contact term being the Einstein–Cartan signature. Variation gives the nonlinear (Hehl–Datta) Dirac equation,

$$i\gamma^{\mu}\partial_{\mu}\psi - m\,\psi \;=\; -\frac{3}{8}\,\kappa\,\big(\bar\psi\,\gamma_5\gamma_a\psi\big)\,\gamma_5\gamma^{a}\psi, \tag{2}$$

whose attractive channel supports self-bound, localized solutions. With the bare gravitational coupling $\kappa = 8\pi G$ the binding scale is Planck-suppressed; the premise of the framework is that the operative coupling is the infrared-amplified one, generated by dimensional transmutation from a propagating spin–torsion sector,

$$\Lambda \;=\; M_{\rm Pl}\,\exp\!\big(-2\pi/b_0\,g^2\big), \tag{3}$$

exactly as $\Lambda_{\rm QCD}$ is generated in the Standard Model. Λ is the single dimensionful scale of the particle sector. Its first-principles value (the coupling g and the one-loop coefficient b_0) is the subject of a separate lattice programme; here we take it as given and measure every mass relative to it. The soliton carries, in addition, a three-valued internal label—a colour forced by the same construction—so that the strong-infrared sector confines, with string tension σ and Sommer radius $r_0 = O(\sigma^{-1/2})$; that structure is developed elsewhere and enters here only as the unit in which the bag is measured.

B. The configurational mass

The condensate the soliton digs is a position-dependent dynamical (constituent) mass $M(r)$, generated by the chiral order parameter in the Nambu–Jona-Lasinio sense [9, 10]. A level bound in this core acquires its rest mass as the field energy stored in the overlap of its density with $M(r)$,

$$m_n \;=\; c_T\!\int\!{\rm d}^3r\;\big|\psi_n(\mathbf{r})\big|^2\,M(r), \tag{4}$$

with c_T a per-sector coupling—the level's grip on the condensate, fixed by its colour and electric charge. Writing the core as $M(r) = M_0\,f(r/R_0)$ for a bag of half-density radius R_0 and fixed profile f, and normalizing the level, the mass becomes Λ times a dimensionless overlap,

$$\begin{aligned} m(T,n) &\;=\; \Lambda\,c_T\,\big|O_n(s_T)\big|^2,\\ O_n(s_T) &\;=\; \frac{\displaystyle\int_0^{\infty}{\rm d}r\;\big|u_n(r)\big|^2 f(r/R_0)}{\displaystyle\int_0^{\infty}{\rm d}r\;\big|u_n(r)\big|^2}, \end{aligned} \tag{5}$$

where $u_n(r)$ is the n-th radial level of the confining well (the torsiton's bag), $n = 1,2,3$ carrying $0,1,2$ interior nodes—the three generations—and

$$s_T \;\equiv\; R_0/r_0 \tag{6}$$

is the bag *sharpness* in units of the confinement scale. We take the mass-giving core to be a Woods–Saxon profile,

$$f(x) \;=\; \frac{1}{1+\exp\!\big[(x-1)/\tau\big]},\qquad x = r/R_0, \tag{7}$$

which falls to one-half at $x = 1$—that is, at $r = R_0$, the half-density radius the lattice measures directly (Sec. III C)—over a surface thickness set by the diffuseness $\tau \ll 1$. The radial levels $u_n(r)$ are the lowest three bound states of the confining well, each normalized by the denominator of Eq. (5). Equations (5)–(7) are the crux: one tower of fitted Yukawa couplings is replaced by one geometric function of one number. A sharp core (small s_T) confines f to the innermost region where only the nodeless $n = 1$ level has appreciable support, suppressing the higher generations; a broad core overlaps all three nearly equally—the lever made precise in Sec. II C.

C. The hierarchy as a lever

The generations differ only in n. The ground level ($n = 1$, nodeless) is the most localized and overlaps the core most; each higher level is more diffuse and node-bearing, so its overlap is progressively smaller. When the core is *sharp* (small s_T) the suppression compounds and the consecutive overlaps form a steeply decreasing geometric tower; when the core is broad the levels weigh nearly alike. The span across the three generations,

$$R(s_T) \;\equiv\; \frac{m(T,3)}{m(T,1)} \;=\; \frac{\big|O_3(s_T)\big|^2}{\big|O_1(s_T)\big|^2}, \tag{8}$$

is therefore an exponentially sensitive function of the bag sharpness. Numerically, in the physically relevant region,

$$\frac{{\rm d}\ln R}{{\rm d}\ln s_T} \;\approx\; 15, \tag{9}$$

so that a few-percent change in s_T moves the span by a factor of a few. The observed charged-lepton span,

$m_\tau/m_e \approx 3477$, is reproduced by the consecutive ladder $n = 1, 2, 3$ for s_T in the *productive window*

$$s_T \;\in\; [0.43,\, 0.70]\, r_0, \tag{10}$$

with the electron's anomalously small mass emerging as the exp(−overlap) tail of the sharpest core rather than a tuned input (Fig. 1). The neutrino's lightness follows in the same scheme from a vanishing c_T: colourless and electrically neutral, it has the weakest grip on the condensate, and needs no new scale [12]. The entire charged-lepton hierarchy thus reduces to the single quantity s_T—which, by Eq. (6), has a clean non-perturbative definition the lattice can measure.

III. LATTICE MEASUREMENT

A. Ensemble and scale

We work on a dynamical $N_f = 2$ SU(3) *fundamental* Wilson ensemble [17]—the fermion determinant in the path integral, so the vacuum is the real chiral-symmetry-breaking condensate, the "weirder than mean field" vacuum in which the physical torsiton and any excited rungs live. The geometry is $16^3 \times 48$ at $\beta = 5.6$, sea mass -0.5, with 243 thermalized, decorrelated configurations. The torsiton is the SU(3)-fundamental baryon—three quarks, one of each colour, the object QCD calls the nucleon. The scale, from the static quark–antiquark potential via the Sommer radius [18], is $r_0/a = 3.13$, with $\sqrt{\sigma}\, a = 0.385$.

B. Binding

The torsiton binds, with a clean effective-mass plateau at $m_N > m_\pi > 0$:

$$\begin{aligned} m_\pi\, a &= 1.269(3), \\ m_N\, a &= 2.091(7), \\ m_N/m_\pi &= 1.65. \end{aligned} \tag{11}$$

At heavy quark mass this ratio approaches the constituent-counting value $m_N/m_\pi \to 3/2$ (a baryon is three quarks, a pion two)—the non-perturbative confirmation that the object binds at the right mass relative to its own confinement scale (Fig. 2). The pion mass-squared is linear in the bare quark mass (Gell-Mann–Oakes–Renner [19]), fixing the chiral trend along which the bag is followed.

C. The bag, by three independent routes

The hierarchy reduces to the sharpness $s_T = R_0/r_0$ of the mass-giving bag, where R_0 is its half-density radius (Eq. 6). We determine it three ways that need not have agreed (Fig. 4):

1. the *one-body proxy*—the gauge-invariant dressed-quark scalar profile $\rho(r) = \mathrm{Tr}[S^\dagger S]$—whose chiral extrapolation gives $s_T \approx 0.43$;
2. the *genuine three-body condensate* $\langle N|\bar{q}q(r)|N\rangle$, a connected nucleon scalar three-point function evaluated with a sequential source and validated by exact reconstruction of the two-point function. Its bag *resolves*: the half-density radius lifts clean off the lattice cutoff, $R_0 = 1.4$–$1.8\, a$ across sink times (against the unresolved $\sim a$ of a coarse pilot), giving $s_T \approx 0.45$–0.56 (Fig. 3), with a stable scalar charge $g_S = 0.21$;
3. the *lepton hierarchy* itself—inverting the lever (Eq. 9) for the observed span requires $s_T \approx 0.43$.

The three cluster at $s_T \approx 0.43$–0.50, within the productive window of Eq. (10). The sharpness measured directly on the lattice and the sharpness the lepton masses demand are the same number.

IV. DISCUSSION

The central finding is a convergence: the bag sharpness measured directly from the lattice condensate agrees, to within the present uncertainties, with the value the charged-lepton span independently requires of the lever (Eq. 9). Where the Standard Model inserts nine masses by hand, one geometric number—resolved off the cutoff and lying in the productive window—organizes the charged-lepton tower, and the same construction drives the neutrino to the floor. The torsiton binds, in the dynamical vacuum, at the right mass relative to its own confinement scale; the mechanism by which a single scale sets the hierarchy is established.

Several features bound how far the present ensemble can be pushed, and point to the next steps. The sea quark is heavy ($m_\pi\, a \approx 1.27$, far from the chiral point), so the three-body sharpness is *bracketed* by a sink-time systematic, $s_T = 0.45$–0.56, rather than pinned; and because the lever of Eq. (9) is steep, that spread dominates the inferred spectrum. The result is therefore a statement about the *mechanism*—the bag is resolved and in the window—rather than a determination of the spectrum to a fixed precision. The steepness itself, however, is not a defect but the content: a single mechanism that turns a modest variation of one geometric quantity into a $\times 3477$ span is exactly what an account of the hierarchy from one number must look like. Pinning s_T to the percent level the lever demands requires the unitary chiral extrapolation (lighter sea masses, with the larger temporal extent needed to stabilize the two-state fit), renormalization of the scalar density, and a continuum limit. The absolute scale Λ of Eq. (3)— equivalently the propagating-torsion β-function—is a separate and harder lattice calculation; until it is in hand the spectrum is relative, anchored to a single measured mass.

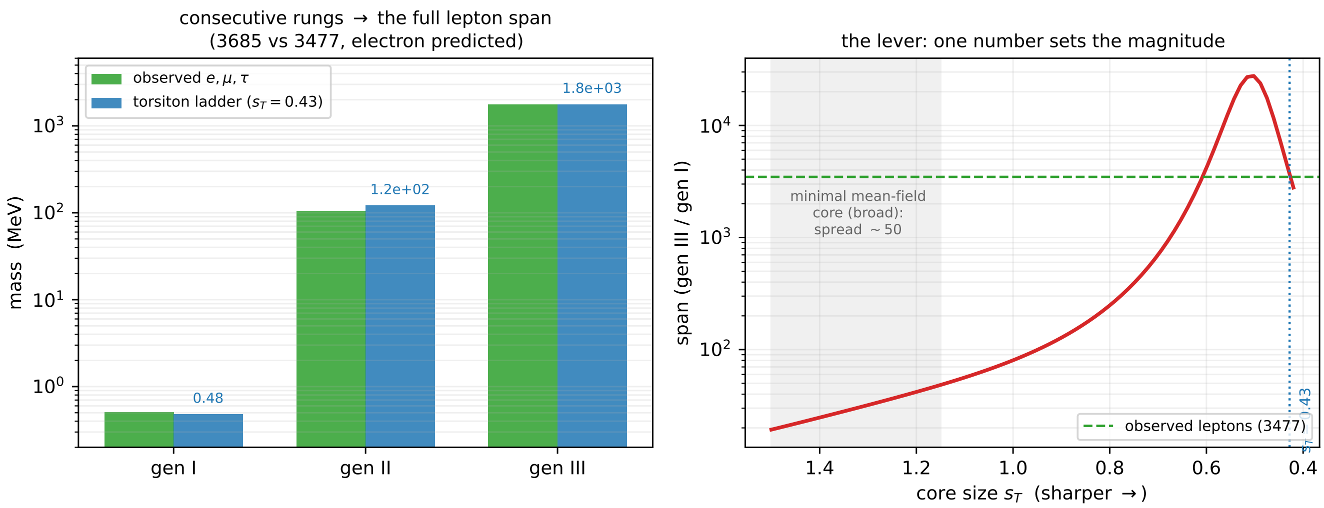

FIG. 1. The hierarchy as one number. *Left:* the consecutive radial ladder $n = 1, 2, 3$ against a core of a single sharpness s_T reproduces the whole charged-lepton span, with the electron's tiny mass a prediction once the heaviest rung sets the scale. *Right:* the lever (Eq. 9)—the span sweeps orders of magnitude as the core sharpens; the observed spread is crossed at a sharp-but-physical core.

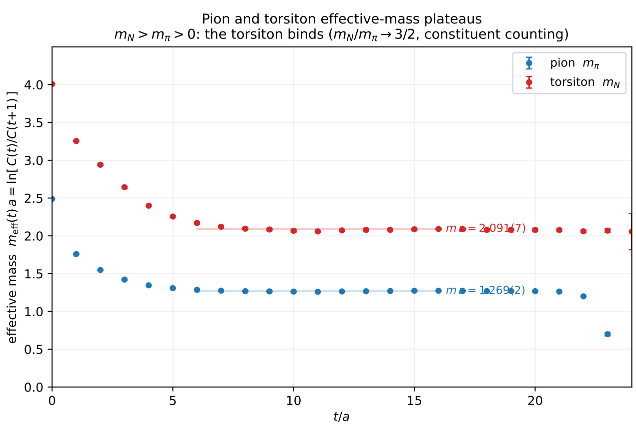

FIG. 2. The pion and torsiton (nucleon) effective-mass plateaus on the dynamical ensemble, $m_{\rm eff}(t)\,a = \ln[\,C(t)/C(t{+}1)\,]$ with jackknife errors over configurations and the plateau-fit bands. A clean plateau at $m_N > m_\pi > 0$ is the torsiton, bound non-perturbatively; the ratio approaches the constituent-counting value $m_N/m_\pi \to 3/2$ at heavy quark mass.

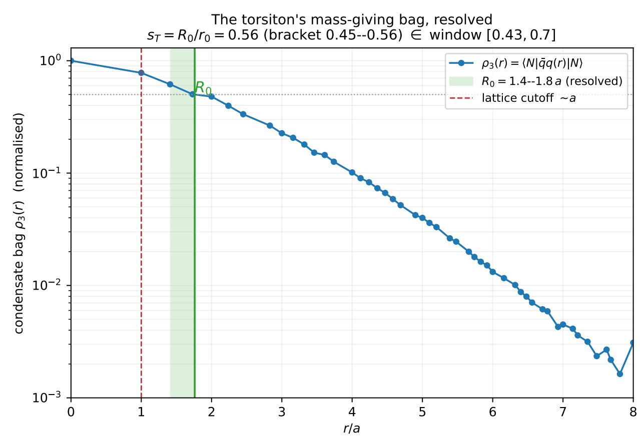

FIG. 3. The torsiton's mass-giving bag, resolved. The connected three-body condensate $\rho_3(r) = \langle N|\bar{q}q(r)|N\rangle$ on the dynamical ensemble ($t_{\rm snk} = 20$). The half-density radius R_0 lifts off the lattice cutoff ($R_0 = 1.4$–$1.8\,a$ across sink times), giving $s_T = R_0/r_0$ inside the productive window $[0.43, 0.70]$.

The torsiton does not stand alone. It is a standing knot of the same gravity–torsion field whose bounce makes the cosmology, and the confining colour structure that sets the unit r_0 used here is the strong-infrared face of that one field. In the broader framework the genesis of all Standard-Model matter proceeds from a single chiral condensation—the coloured sector confined into the baryon octet and decuplet, the colourless sector left as the lepton tower measured above—rather than the staged electroweak-then-QCD sequence, and the electroweak sector is composite, contingent on a near-conformal regime that keeps the Peskin–Takeuchi S small. These developments, and the relation $\sigma = 2\pi v^2$ tying the confinement scale to the condensate, are presented in the companion monograph [20]; the present result supplies the lattice anchor for the lepton sector of that programme.

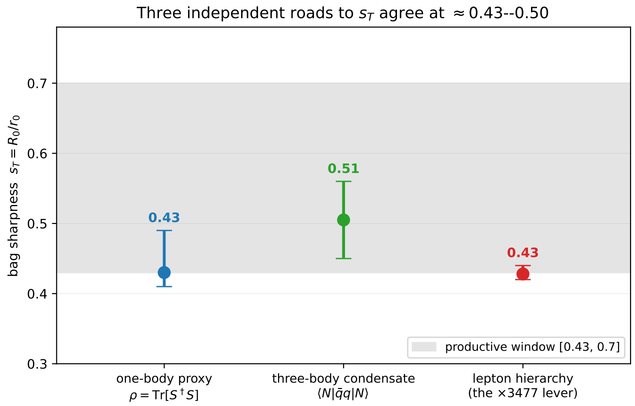

FIG. 4. Three independent roads to the bag sharpness s_T—the one-body dressed-quark profile, the genuine three-body condensate, and the value the observed charged-lepton span requires of the lever (Eq. 9). They agree at $s_T \approx 0.43$–0.50 (shaded: the productive window). The three-body value is bracketed by a sink-time systematic.

V. CONCLUSION

We have argued that the elementary fermions are torsitons—chiral solitons of the nonlinear Einstein–Cartan–Dirac equation—whose mass is configurational, and that the charged-lepton hierarchy reduces to a single quantity, the sharpness $s_T = R_0/r_0$ of the soliton's mass-giving core. On a dynamical $N_f = 2$ SU(3) lattice the torsiton binds at the constituent-counting mass, and its core is resolved by the genuine three-body condensate, landing at $s_T \approx 0.45$–0.56—inside the productive window that the observed $\times 3477$ span requires, and in agreement with both a one-body proxy and the value the lepton masses themselves demand. One geometric number, lattice-measurable, where the Standard Model inserts nine. The chiral-limit value of s_T and the absolute scale Λ are the next, decisive steps; the former either holds the window or falsifies the mechanism.

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